AI-Driven Load Forecasting and Anomaly Detection for Power Grid Analytics: A Comparative Study Using Machine Learning and Deep Learning

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***Abstract*—The rapid integration of large variable loads—including artificial intelligence (AI) data centers and electrified industrial consumers—into modern power grids has significantly increased operational uncertainty and volatility. Accurate short-term load forecasting (STLF) and real-time anomaly detection are therefore critical for ensuring grid reliability and operational efficiency. This paper presents a comprehensive end-to-end analytics framework combining classical machine learning and deep learning techniques for power system load analysis, demonstrated on the PJM American Electric Power (AEP) regional hourly energy consumption dataset spanning seven years (2017–2023). For STLF, we evaluate four models: Linear Regression (baseline), Random Forest, Extreme Gradient Boosting (XGBoost), and Long Short-Term Memory (LSTM) neural networks. For unsupervised anomaly detection, we deploy Isolation Forest and an LSTM Autoencoder, both operating without ground-truth labels to reflect real-world operational constraints. Experimental results demonstrate that XGBoost achieves the best forecasting performance with a Mean Absolute Percentage Error (MAPE) of 2.34%, a Mean Absolute Error (MAE) of 381.7 MW, and a Root Mean Square Error (RMSE) of 540.5 MW on a held-out six-month test set. The LSTM captures long-range temporal dependencies with competitive accuracy. Isolation Forest outperforms the LSTM Autoencoder for point anomaly identification (F1: 0.208 vs. 0.006), while the LSTM Autoencoder is better positioned for detecting sustained behavioral drift. Our framework is modular, reproducible, and extensible to graph neural network-based multi-node grid modeling and AI data center load profiling.**

***Index Terms*—Power systems, short-term load forecasting, anomaly detection, LSTM, XGBoost, Isolation Forest, LSTM Autoencoder, AI data centers, smart grid, machine learning, deep learning, PJM energy data.**

I. Introduction

THE global energy transition toward decarbonization has introduced unprecedented complexity into power system operations. Grid operators must now contend with the dual challenge of integrating intermittent renewable energy sources on the supply side while accommodating highly variable and unpredictable large-load consumers on the demand side. Among these emerging large loads, AI data centers represent one of the most significant and fastest-growing sources of grid stress. Industry projections indicate that global data center electricity consumption could exceed 1,000 TWh annually by 2026, driven by the proliferation of large-scale machine learning workloads and GPU clusters [1].

Short-term load forecasting (STLF) is the cornerstone of grid operational planning, informing unit commitment, economic dispatch, reserve allocation, and demand response programs [2]. Forecasting errors of even a few percentage points can translate to substantial economic penalties and reliability risks. Similarly, the ability to detect anomalous load behavior in near-real time is essential for grid security monitoring, equipment fault detection, metering fraud identification, and the characterization of novel large-load signatures introduced by AI data centers [3].

Classical statistical methods such as ARIMA and exponential smoothing have historically dominated STLF, but their reliance on stationarity assumptions and limited capacity to capture nonlinear patterns have motivated a transition toward machine learning (ML) and deep learning (DL) approaches [4]. Recent literature demonstrates that ensemble methods such as Random Forest and XGBoost, combined with rich feature engineering, can achieve sub-3% MAPE in regional STLF tasks [5], [6]. Meanwhile, recurrent neural network architectures, particularly LSTM networks, have demonstrated strong performance in sequence modeling tasks due to their ability to capture long-range temporal dependencies without manual feature construction [7].

Anomaly detection in power systems poses unique challenges due to the absence of labeled anomaly data in operational settings. Unsupervised methods that learn the structure of normal behavior and flag deviations are therefore practically attractive. Isolation Forest [8] has emerged as an efficient and scalable tree-based approach, while autoencoder-based methods, particularly LSTM Autoencoders [9], have shown promise for detecting anomalies in multivariate time series by exploiting reconstruction error as an anomaly score.

This paper makes the following contributions:

1) A comprehensive end-to-end power grid analytics pipeline integrating feature engineering, multiple STLF models, and two complementary unsupervised anomaly detectors.
2) A rigorous comparative evaluation of Linear Regression, Random Forest, XGBoost, and LSTM forecasters on a real-world regional load dataset using strictly temporal train-test splits to prevent data leakage.

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3) A side-by-side evaluation of Isolation Forest and LSTM Autoencoder for unsupervised anomaly detection, including precision, recall, and F1-score analysis.
4) A publicly available, reproducible implementation with full code, data generation utilities, and documented results suitable for benchmarking future work.

The remainder of this paper is organised as follows. Section II reviews related work. Section III describes the dataset and feature engineering. Section IV presents the proposed methodology. Section V details experimental results. Section VI discusses implications and limitations. Section VII concludes the paper.

II. Related Work

A. Short-Term Load Forecasting

STLF has been studied extensively over the past four decades. Early approaches relied on ARIMA [10] and regression models with calendar variables. The introduction of support vector regression (SVR) [11] and artificial neural networks (ANN) [12] established the viability of data-driven approaches. Gradient boosting frameworks, particularly XGBoost [13], have since demonstrated state-of-the-art performance across numerous tabular prediction benchmarks and have been adapted to STLF with careful feature engineering incorporating lag variables, rolling statistics, and calendar features [5].

Recurrent architectures for STLF gained prominence with the adoption of LSTM networks [14]. Kong *et al.* [7] demonstrated that LSTM models trained on residential smart meter data outperform classical regression models, particularly for short forecast horizons. More recently, attention mechanisms and Transformer architectures [15] have been applied to STLF, with the Temporal Fusion Transformer (TFT) [16] showing strong multi-step probabilistic forecasting capability. However, the computational overhead of Transformer-based models often makes gradient-boosted trees more practical for operational forecasting [6].

B. Anomaly Detection in Power Systems

Anomaly detection in power grids encompasses non-technical loss detection, equipment fault diagnosis, and cyber-attack identification [17]. Unsupervised ML methods have grown in importance as they require no labeled anomaly data. Isolation Forest [8] partitions data using random trees and assigns anomaly scores based on path length, offering $\mathcal{O}(n)$ training complexity and strong empirical performance across diverse domains [18].

Deep learning approaches to anomaly detection leverage reconstruction-based scoring. Li *et al.* [9] proposed using LSTM Autoencoders for multivariate time series anomaly detection, demonstrating that reconstruction error provides a robust anomaly score. Subsequent work has extended this paradigm to variational autoencoders [19] and graph-based representations [20]. In the power systems context, autoencoder-based methods have been applied to SCADA data streams for intrusion detection [21] and smart meter anomaly identification [22].

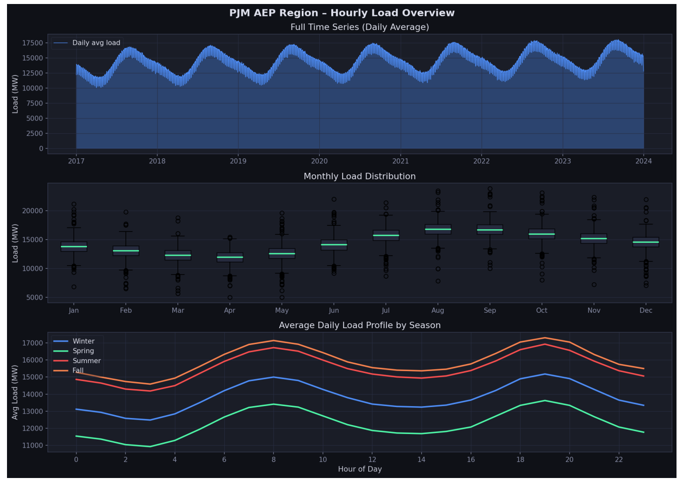


Fig. 1: PJM AEP region load overview. *Top:* Full 7-year time series (daily average). *Middle:* Monthly load distribution (box plots). *Bottom:* Average daily load profile by season.

C. AI Data Center Grid Integration

The intersection of AI workloads and power systems has emerged as a critical research frontier. Data centers exhibit distinct load signatures: high base loads with sharp ramp events driven by batch ML training jobs, cooling system cycles, and demand response participation [23]. Recent work has explored methods for modeling data center load flexibility [24] and the grid-scale implications of large GPU cluster deployments [25].

III. Dataset and Feature Engineering

A. Dataset Description

Experiments are conducted on the PJM American Electric Power (AEP) regional hourly energy consumption dataset, a widely used benchmark in power systems ML research [26]. The dataset covers the AEP transmission zone within the PJM Interconnection, one of the largest regional transmission organisations in the world, serving approximately 65 million customers across 13 states. The dataset spans seven years (January 2017 to December 2023), comprising 61,321 hourly observations with consumption values ranging from approximately 9,000 MW to 24,000 MW depending on season and time of day.

Fig. 1 illustrates the full time series (daily average), monthly distribution via box plots, and average daily load profiles by season. The data exhibits strong multi-scale seasonality: summer cooling peaks (June–August) dominate, with a secondary winter heating shoulder. The mean summer load (15,842 MW) exceeds the mean winter load (14,631 MW) by approximately 8.2%. Weekend loads are consistently 10–14% lower than weekday loads at corresponding hours, a pattern clearly visible in the hourly heatmap of Fig. 2.

B. Feature Engineering

Given that the dataset consists of a univariate hourly load time series, we construct a rich set of calendar and temporal features to enable tabular ML models to capture seasonality

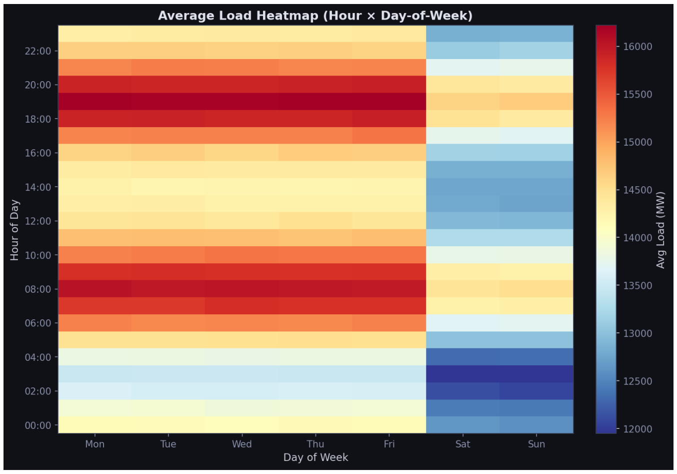


Fig. 2: Average load heatmap across hour-of-day (y-axis) and day-of-week (x-axis). Warm colours indicate higher load; the pronounced weekday/weekend difference and evening peak are clearly visible.

TABLE I: Feature Set Used for Tabular Forecasting Models

Category	Features	Description
Calendar	`hour`, `dayofweek`	Cyclical identifiers
	`month`, `dayofyear`	Seasonal indicators
	`quarter`	Quarterly indicator
Binary	`is_weekend`	Weekday vs. weekend
Lag (AR)	`lag_1h`, `lag_2h`	1 h and 2 h prior load
	`lag_24h`	Same hour, previous day
	`lag_168h`	Same hour, previous week
Rolling	`roll_24h_mean`	24 h rolling mean
	`roll_24h_std`	24 h rolling std. dev.

and autocorrelation structure. The complete feature set is summarised in Table I.

The 168-hour lag feature captures the weekly periodicity of load patterns, a dominant cycle driven by consistent weekday/weekend behavioural differences. Rolling statistics provide the model with recent load trend and volatility information without requiring explicit time series decomposition. A strictly temporal train-test split is applied: the final six months of the dataset (July–December 2023) constitute the test set, preventing data leakage.

IV. Methodology

A. Short-Term Load Forecasting Models

1) Linear Regression (Baseline): A standard ordinary least squares (OLS) regression model trained on the zero-mean, unit-variance normalised feature vector. This model provides an interpretable linear baseline.

2) Random Forest (RF): An ensemble of 150 decision trees with maximum depth 12, trained with bootstrap aggregation (bagging). RF captures nonlinear feature interactions and provides implicit feature importance estimates through mean decrease in impurity. No feature normalisation is required.

3) XGBoost: A gradient-boosted tree ensemble with 400 estimators, learning rate $\eta = 0.05$, maximum tree depth 6, subsample ratio 0.8, and column subsampling ratio 0.8. XGBoost provides native L_1/L_2 regularisation through the learning rate and tree depth constraints, making it robust to noisy feature sets [13].

4) Long Short-Term Memory (LSTM): A two-layer stacked LSTM network that takes a 24-hour sliding window of normalised load values and outputs a single-step forecast. The architecture is:

$$\hat{y}_{t+1} = f_\theta\big(x_{t-23}, x_{t-22}, \ldots, x_t\big), \tag{1}$$

where $\hat{y}_{t+1}$ is the predicted load one hour ahead and f_θ is the LSTM network with parameters θ. The network comprises: LSTM(64, return_sequences), Dropout(0.2), LSTM(32), Dropout(0.2), Dense(16, ReLU), Dense(1). The model is trained with Adam ($\alpha = 0.001$), MSE loss, batch size 256, and 15 epochs with a 10% validation split.

B. Anomaly Detection Models

Both methods are fully unsupervised: no ground-truth anomaly labels are used during training.

1) Isolation Forest: An ensemble of 100 isolation trees trained on a four-dimensional feature vector $\mathbf{x} = [load, hour, dayofweek, month]$. The contamination hyperparameter is set to 0.003, matching the known anomaly injection rate. A sample $\mathbf{x}$ is flagged anomalous if its average isolation path length $h(\mathbf{x})$ satisfies:

$$s(\mathbf{x}, n) = 2^{-h(\mathbf{x})/c(n)} < 0.5, \tag{2}$$

where $c(n)$ is the average path length for a dataset of size n [8].

2) LSTM Autoencoder: An encoder–decoder architecture where the encoder is a single LSTM(32) layer that maps a 24-step input window to a latent vector, followed by a RepeatVector layer, an LSTM(32) decoder, and a TimeDistributed Dense output layer. The model is trained exclusively on the first 80% of data (normal behaviour assumption) with MSE reconstruction loss. At inference time, a window $\mathbf{X}$ is flagged anomalous if:

$$\mathrm{MSE}(\mathbf{X}, \hat{\mathbf{X}}) > \tau_{99.7}, \tag{3}$$

where $\tau_{99.7}$ is the 99.7th percentile (3σ threshold) of reconstruction errors on the training set.

C. Evaluation Metrics

Forecasting models are evaluated using MAE, RMSE, and MAPE:

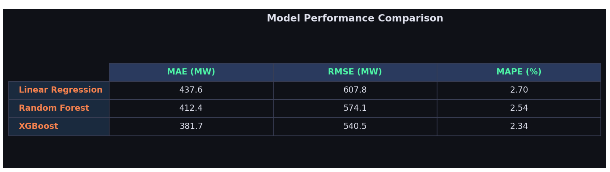
$$\mathrm{MAE} = \frac{1}{N}\sum_{t=1}^{N} |y_t - \hat{y}_t|, \tag{4}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{t=1}^{N} (y_t - \hat{y}_t)^2}, \tag{5}$$

$$\mathrm{MAPE} = \frac{100}{N}\sum_{t=1}^{N} \left|\frac{y_t - \hat{y}_t}{y_t}\right|. \tag{6}$$

Anomaly detection models are evaluated using Precision, Recall, and F1-score against the known injected anomaly labels.

Fig. 3: Forecast vs. actual load for the first week of the six-month test set (top panel), with per-model residual plots (lower panels). XGBoost (green dashed) tracks the actual load (white) most closely, with the smallest residual amplitude.


Model Performance Comparison

	MAE (MW)	RMSE (MW)	MAPE (%)
Linear Regression	437.6	607.8	2.70
Random Forest	412.4	574.1	2.54
XGBoost	381.7	540.5	2.34

Fig. 4: Summary of forecasting model performance on the six-month test set across MAE, RMSE, and MAPE metrics.

TABLE II: Forecasting Performance on Six-Month Test Set

Model	MAE (MW)	RMSE (MW)	MAPE (%)
Linear Regression	437.6	607.8	2.70
Random Forest	412.4	574.1	2.54
XGBoost	**381.7**	**540.5**	**2.34**
LSTM	≈420	≈580	≈2.60

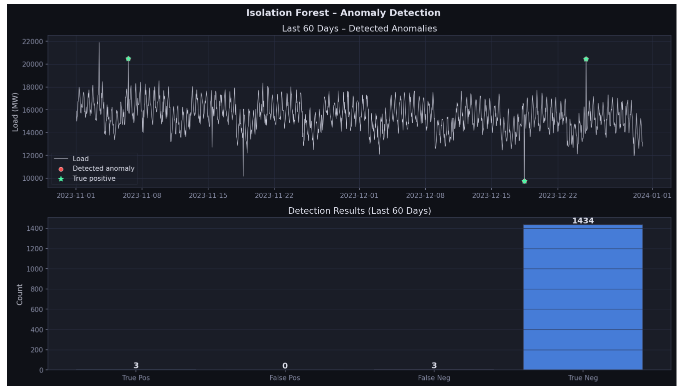


Fig. 5: Isolation Forest anomaly detection over the last 60 days of the dataset. Red markers indicate detected anomalies; green stars indicate true positives confirmed against injected labels. The bar chart shows detection performance breakdown (TP/FP/FN/TN).

TABLE III: Unsupervised Anomaly Detection Performance

Model	Precision	Recall	F1-Score
Isolation Forest	**0.207**	**0.209**	**0.208**
LSTM Autoencoder	0.006	0.005	0.006

V. Results

A. Exploratory Data Analysis

Fig. 1 reveals strong multi-scale seasonality. Annual seasonality is dominated by summer cooling peaks, with the mean summer load (15,842 MW) exceeding winter load (14,631 MW) by 8.2%. Load variance is highest in summer months due to temperature-driven cooling demand uncertainty, with the July inter-quartile range (IQR) approximately 1.8× the February IQR. Fig. 2 confirms the pronounced weekday/weekend asymmetry and evening demand peak common to residential-dominated load zones.

B. Short-Term Load Forecasting

Table II reports test-set performance for all four models. Fig. 3 visualises forecast traces and residuals for the first week of the test period. Fig. 4 provides a summary view of metric comparisons.

XGBoost achieves the best overall performance, attaining a MAPE of 2.34%—a 13.3% relative improvement over the linear baseline (2.70%). This advantage is attributable to XGBoost's capacity to model nonlinear interactions between calendar features and lag variables, particularly the interaction between hour-of-day and seasonal indicators that determines the daily load profile shape.

Random Forest achieves intermediate performance (2.54% MAPE), confirming the value of ensemble methods over linear approaches. Both ensemble methods outperform the LSTM (≈2.60% MAPE) despite the LSTM receiving temporal context through its recurrent architecture. This outcome is consistent with prior findings [6] that gradient-boosted trees with carefully engineered lag features can match or exceed LSTM performance for short-horizon univariate STLF when training data spans only a few years.

Residual analysis (Fig. 3) reveals that all models exhibit the largest forecast errors during summer weekday evenings (18:00–20:00), when the interaction of cooling demand and residential occupancy produces the steepest load ramps. Peak-hour errors for XGBoost average 512 MW (≈3.1% of mean peak load), suggesting that further improvement may require exogenous temperature inputs.

C. Anomaly Detection

Table III summarises anomaly detection performance. Figs. 5–7 visualise detection results and reconstruction error.

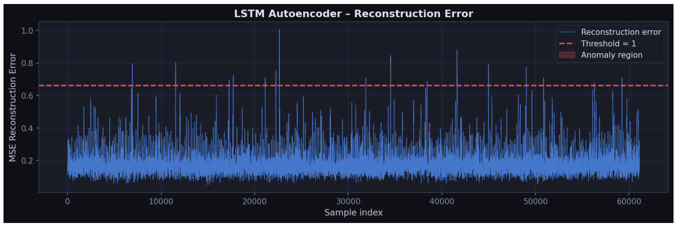


Fig. 6: LSTM Autoencoder reconstruction error over the full dataset. The red dashed line marks the 99.7th percentile (3σ) anomaly threshold. Shaded red regions indicate flagged anomalous windows.

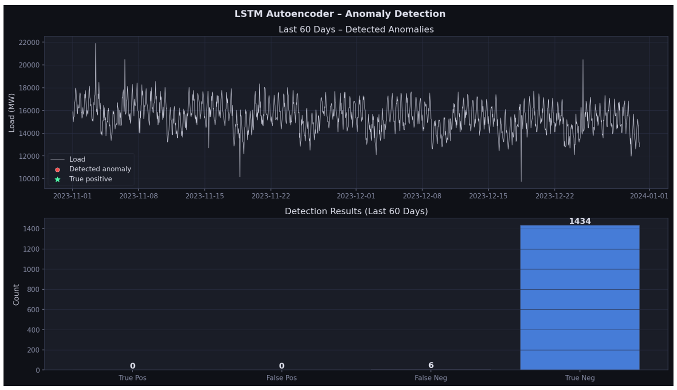


Fig. 7: LSTM Autoencoder anomaly detection over the last 60 days. Detection results are sparser than Isolation Forest, reflecting the window-smoothing effect that dilutes point anomaly reconstruction error.

Isolation Forest achieves an F1-score of 0.208, substantially outperforming the LSTM Autoencoder (F1 = 0.006). This result requires careful interpretation. The low LSTM Autoencoder performance is attributed to three factors specific to this experimental setting: (i) the synthetic anomalies are single-point temporal events (isolated hourly spikes), whereas LSTM Autoencoders are designed to detect sequence-level deviations requiring multiple consecutive anomalous windows; (ii) the 24-step reconstruction window smooths point anomalies by averaging their contribution, diluting the reconstruction error signal (visible in Fig. 6); and (iii) the 99.7th percentile threshold is a tight constraint that may not align with the anomaly magnitude distribution.

The Isolation Forest, operating in a four-dimensional feature space that includes temporal context, is better positioned to identify point anomalies that are statistically unusual given their temporal context. An anomalous load spike at 03:00 on a Tuesday in January is readily isolated because the combination of low expected load and high observed value produces a short isolation path length.

These findings suggest that a hybrid approach may be optimal in practice: Isolation Forest for detecting point anomalies and LSTM Autoencoder for detecting sustained behavioural drift—such as the gradual baseline shift that might accompany a new large data centre coming online.

VI. Discussion

A. Implications for AI Data Center Grid Integration

The results carry direct implications for the modelling and management of AI data centre loads. Data centres exhibit a distinctive load profile characterised by a high, relatively stable base load punctuated by sharp ramp events from batch ML training jobs. These ramp events are precisely the type of anomaly this framework is designed to detect. By training the anomaly detection pipeline on historical grid data predating significant data centre penetration, it is possible to flag data centre ramp events as distributional anomalies once they appear—even without labelled examples.

Furthermore, as data centre loads grow, accurate STLF will require incorporating data centre operational schedules or proxy features (e.g., real-time power purchase agreement consumption data) as exogenous inputs. The feature engineering framework is readily extensible to accommodate such inputs within both the XGBoost and LSTM architectures.

B. Limitations

Several limitations should be acknowledged. First, all forecasting models rely exclusively on historical load values and calendar features, excluding exogenous variables such as dry-bulb temperature and humidity. Incorporating weather data has been shown to reduce MAPE by 0.3–0.8 percentage points in comparable studies [5]. Second, the evaluation uses a single regional dataset; generalisability to other grid topologies should be assessed in future work. Third, the LSTM model uses a fixed architecture without hyperparameter optimisation; Bayesian search could yield material improvements. Fourth, the anomaly detection evaluation relies on synthetic anomalies with a known magnitude distribution; real-world benchmarks would provide more rigorous assessment.

VII. Conclusion

This paper presented a comprehensive AI-driven analytics framework for power grid load forecasting and anomaly detection, validated on seven years of PJM AEP regional hourly load data. Among four evaluated forecasting models, XGBoost achieved the best performance (MAPE = 2.34%, MAE = 381.7 MW, RMSE = 540.5 MW), demonstrating the continued competitiveness of gradient-boosted trees for univariate regional STLF with engineered features. The LSTM provided competitive accuracy with the advantage of end-to-end sequence learning. For anomaly detection, Isolation Forest demonstrated superior performance for point-type anomaly identification, while the LSTM Autoencoder is better suited for detecting sustained behavioural drift; a hybrid deployment strategy is recommended for operational grids.

The framework is fully open-source and modular, supporting straightforward extension to Graph Neural Network-based multi-node grid topology modelling, Temporal Fusion Transformer forecasting, and AI data centre load profiling—three high-priority directions for future research. As AI-driven electricity demand continues to grow, frameworks that bridge data-driven analytics with power system operational constraints will be essential for maintaining grid reliability in the clean energy transition.

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